

Module 3: IAM Roles Assignment

- *By MG Ishaque*

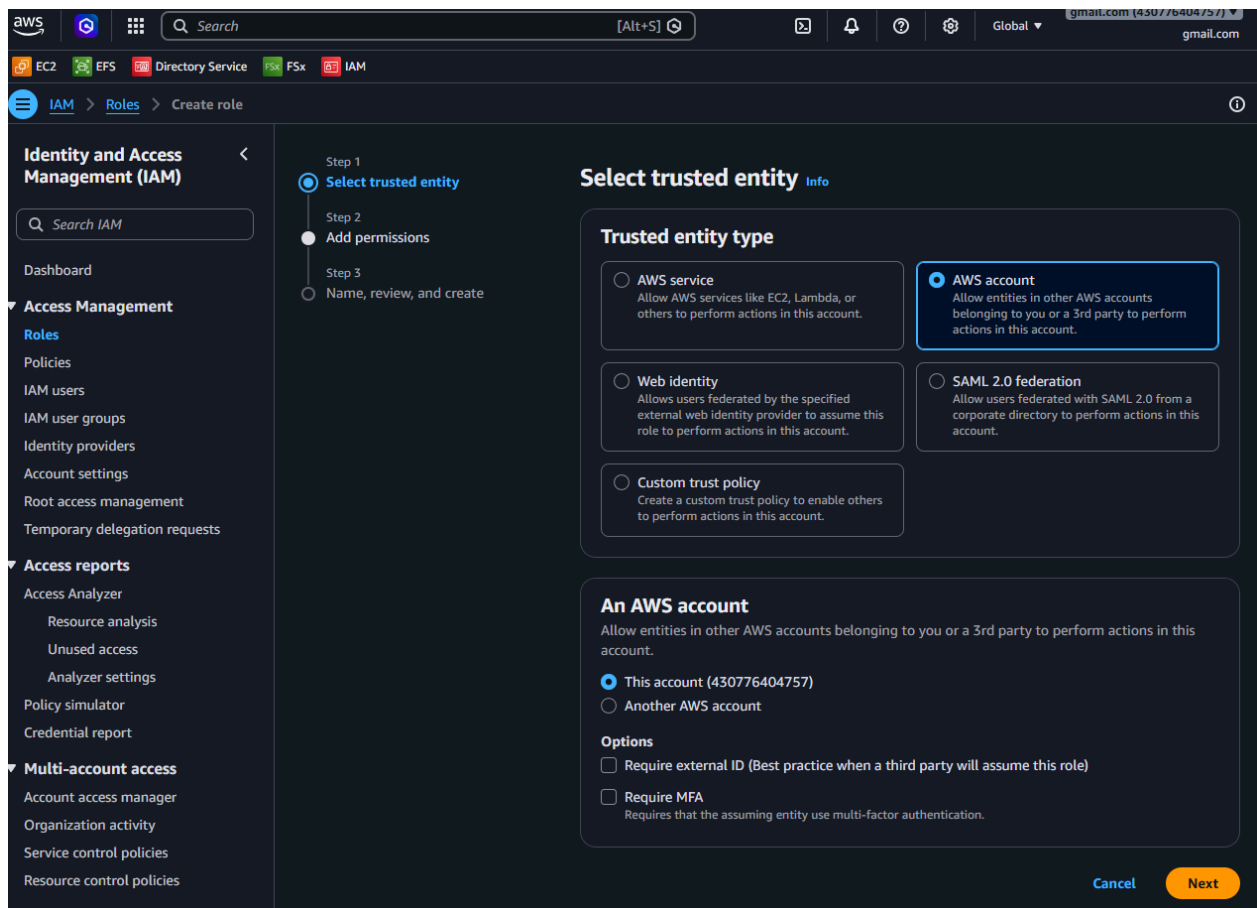
Problem Statement: “ You work for XYZ Corporation. To maintain the security of the AWS account and the resources you have been asked to implement a solution that can help easily recognize and monitor the different users. Tasks To Be Performed: 1. Create a role which only lets user1 and user2 from task 1 to have complete access to VPCs and DynamoDB. 2. Login into user1 and shift to the role to test out the feature.”

Tasks Performed:

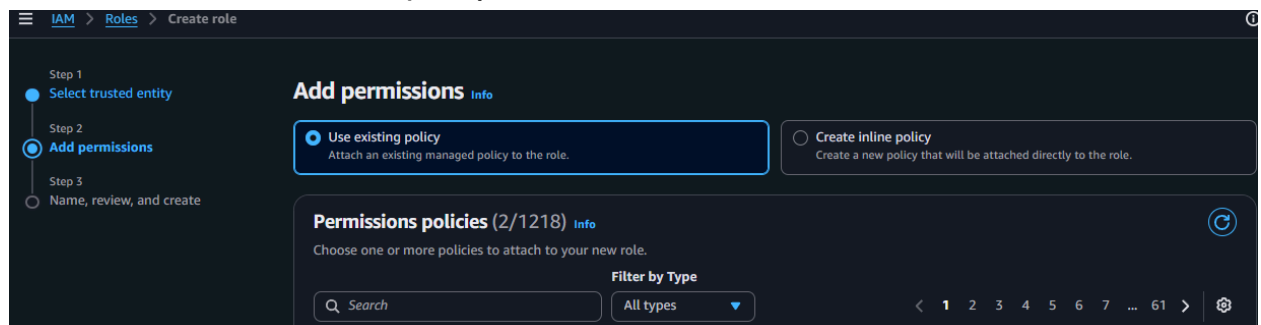
1. Created a role which only lets users Dev1 and Dev2 from task 1 to have complete access to VPCs and DynamoDB.
2. Logged in into Dev1 and shifted to the role to test out the feature.

Task 1: Created the IAM Role

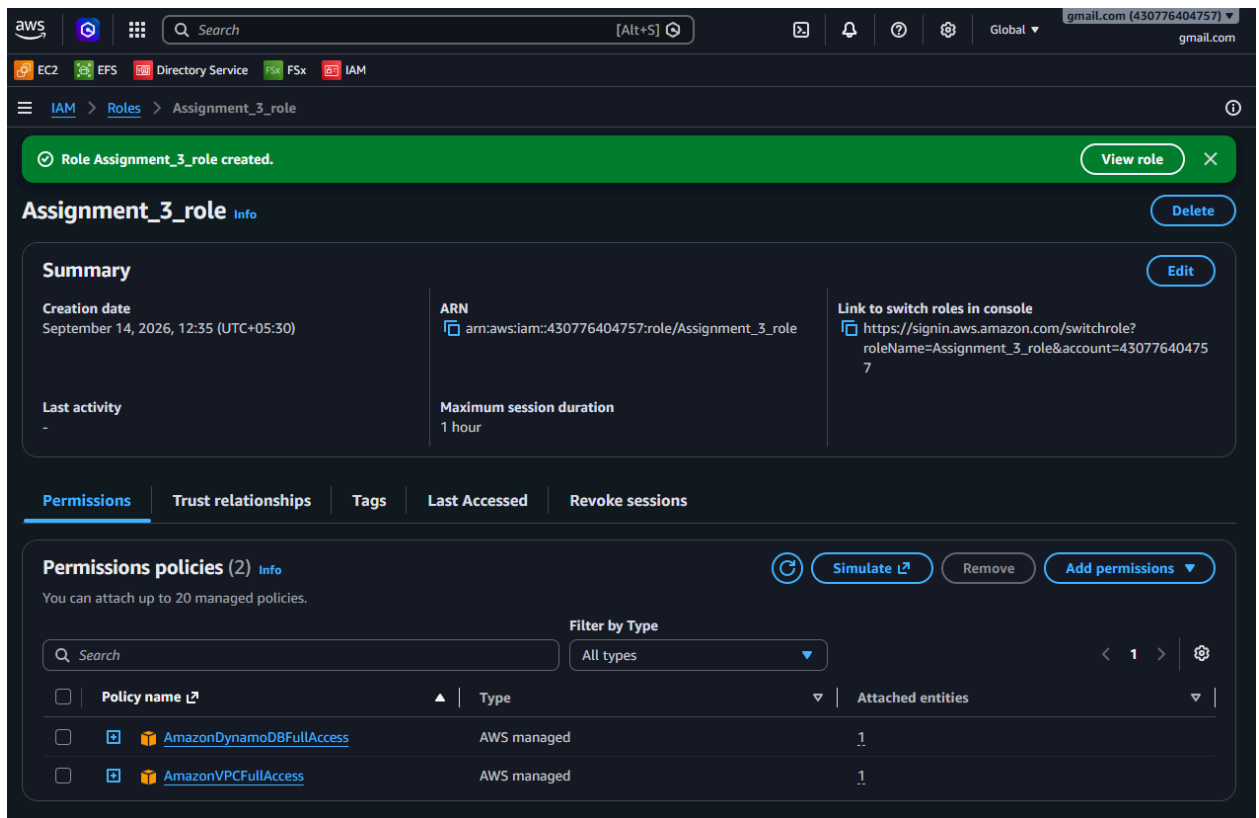
1. The **IAM Console** was opened.
2. From the left navigation menu, **Roles** was selected.
3. Clicked **Create role**.
4. Under Trusted entity type: **AWS account** was selected.
5. **This account** was selected as the trusted AWS account.



6. The **AmazonDynamoDBFullAccess** policy was selected.
7. The **AmazonVPCFullAccess** policy was selected.

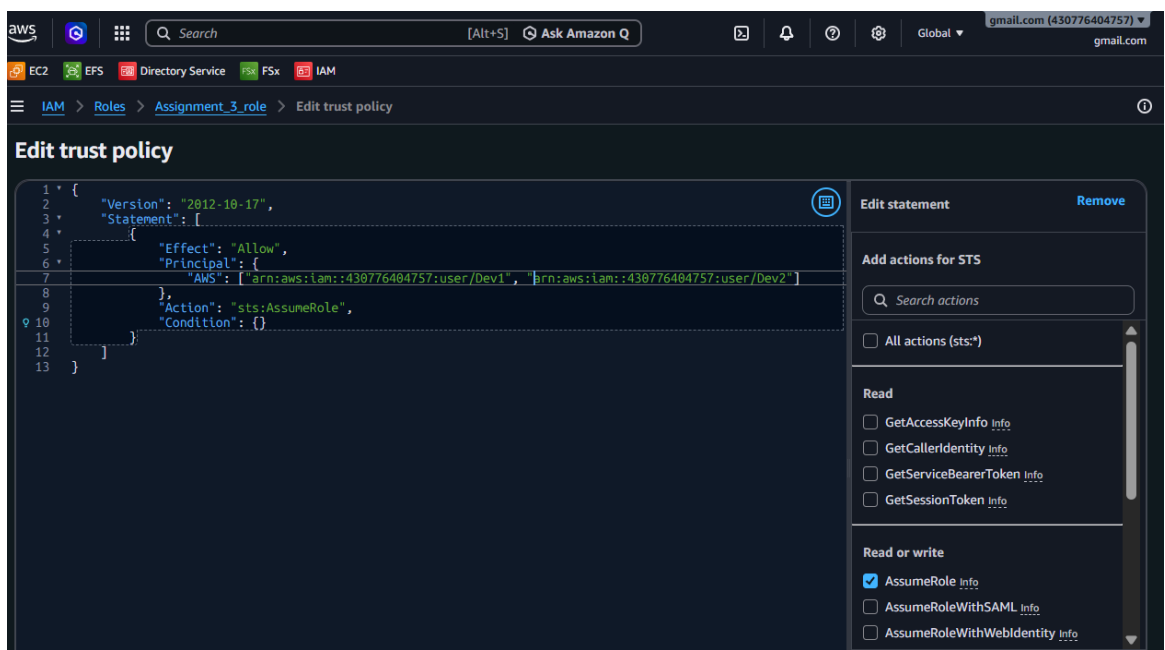


8. The role name **Assignment_3_role** was entered.
9. The role was created successfully.



Task 2: Restricted the Role to Dev1 and Dev2

13. **Assignment_3_role** was opened.
14. The **Trust relationships** tab was selected.
15. **Edit trust policy** was clicked.
16. The trust policy was updated so that only **Dev1** and **Dev2** were allowed to assume the role.



The trust policy used was:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": [
          "arn:aws:iam::430776404757:user/Dev1",
          "arn:aws:iam::430776404757:user/Dev2"
        ]
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

17. Update policy was clicked.

The screenshot shows the AWS IAM console interface. At the top, a green banner indicates "Trust policy updated." The main heading is "Assignment_3_role" with an "Info" link and "Delete" and "Edit" buttons. Below this is a "Summary" section with a table containing the following information:

Summary	
Creation date September 14, 2026, 12:35 (UTC+05:30)	ARN arn:aws:iam::430776404757:role/Assignment_3_role
Last activity -	Maximum session duration 1 hour

Below the summary is a "Link to switch roles in console" with the URL: https://signin.aws.amazon.com/switchrole?roleName=Assignment_3_role&account=430776404757

The "Trust relationships" tab is selected, showing a list of trusted entities. The trust policy JSON is displayed in a code editor:

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": [
8           "arn:aws:iam::430776404757:user/Dev1",
9           "arn:aws:iam::430776404757:user/Dev2"
10        ]
11      },
12      "Action": "sts:AssumeRole",
13      "Condition": {}
14    }
15  ]
16 }
```

Task 3: Logged in as Dev1

18. The IAM user sign-in page was opened.
19. The account ID 430776404757 was entered.
20. The IAM username Dev1 was entered.
21. The password was entered.
22. Sign in was clicked.

The screenshot shows the AWS IAM user sign-in page on the left and the Amazon Lightsail banner on the right. The sign-in page has the following fields and options:

- IAM user sign in** (with an information icon)
- Account ID or alias ([Don't have?](#)): 430776404757
- ☐ Remember this account
- IAM username: Dev1
- Password: [masked]
- ☐ Show Password [Having trouble?](#)
- Sign in** (orange button)
- [Sign in using root user email](#)
- [Create a new AWS account](#)

The Amazon Lightsail banner on the right features the text "Amazon Lightsail" and "Lightsail is the easiest way to get started on AWS", with a "Learn more »" button and a robot character.

23. The AWS account menu was opened from the top-right corner.
Switch role was selected.

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the account ID 4307-7640-4757. The main content area is titled "Console Home" and displays a "Recently visited" section with a message "No recently visited services" and a "Welcome to AWS" section with links to "Getting started with AWS" and "Training and certification". On the right side, the account menu is open, showing the following options:

- Account ID: 4307-7640-4757
- Account name: Access denied
- Account color: Access denied
- IAM user: Dev1
- Account
- Organization
- Service Quotas
- Billing and Cost Management
- Security credentials
- Console Mobile App
- Agent Toolkit for AWS
- Turn on multi-session support** (button)
- Switch role** (button)
- Sign out** (button)

24. The account ID **430776404757** was entered.
25. The IAM role name **Assignment_3_role** was entered.
26. **Switch Role** was clicked.
27. The AWS console displayed **Assignment_3_role @ 430776404757**.

Switch Role

Switching roles enables you to manage resources across Amazon Web Services accounts using a single user. When you switch roles, you temporarily take on the permissions assigned to the new role. When you exit the role, you give up those permissions and get your original permissions back. [Learn more](#)

Account ID
The 12-digit account number or the alias of the account in which the role exists.

430776404757

IAM role name
The name of the role that you want to assume which can be found at the end of the role's ARN. For example, provide the **TestRole** role name from the following role ARN: `arn:aws:iam::123456789012:role/TestRole`.

Assignment_3_role

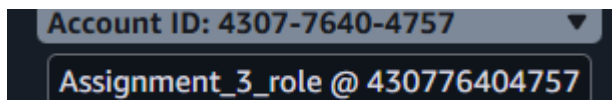
Display name - optional
This name will appear in the console navigation bar when active. Choose a name to help identify the permission set assigned to the role.

Assignment_3_role @ 430776404757

Display color - optional
The selected color displays in the console navigation when this role is active

☐ None

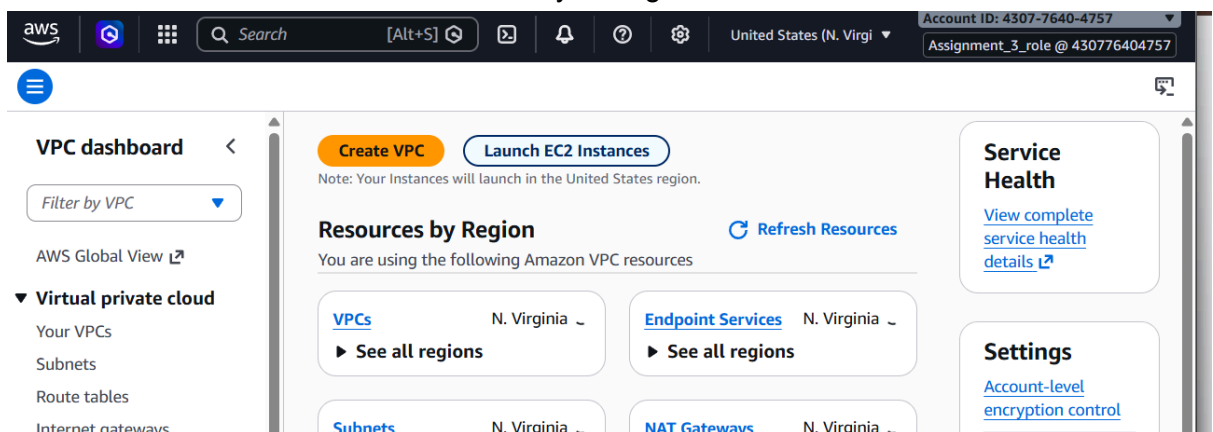
[Cancel](#) [Switch Role](#)



Task 4: Tested the Permissions

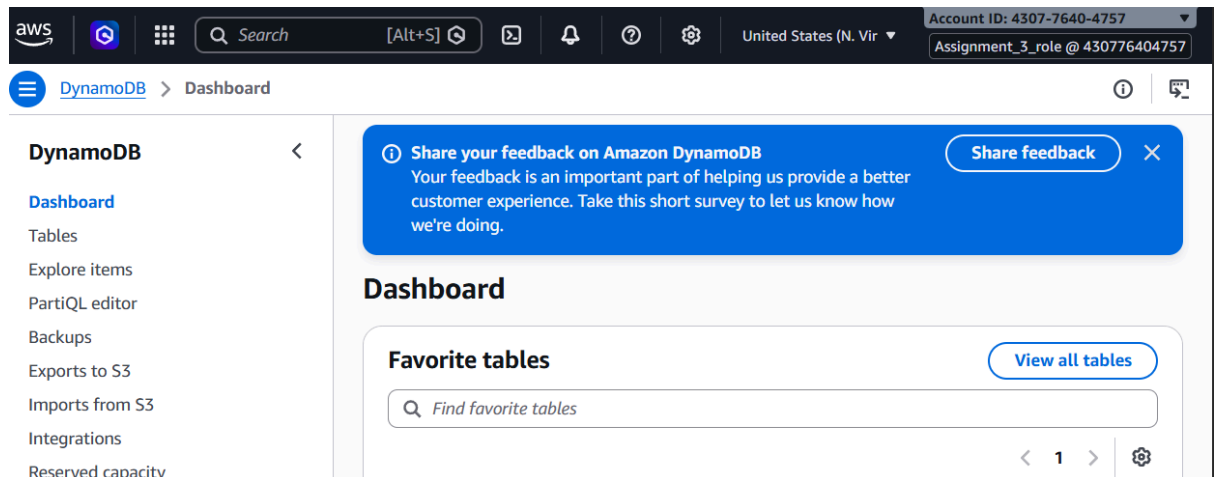
Verified Allowed Access

28. The **VPC Dashboard** was opened.
29. VPC resources were accessed successfully through the assumed role.



30. **DynamoDB** was opened.

31. The DynamoDB dashboard was accessed successfully through the assumed role

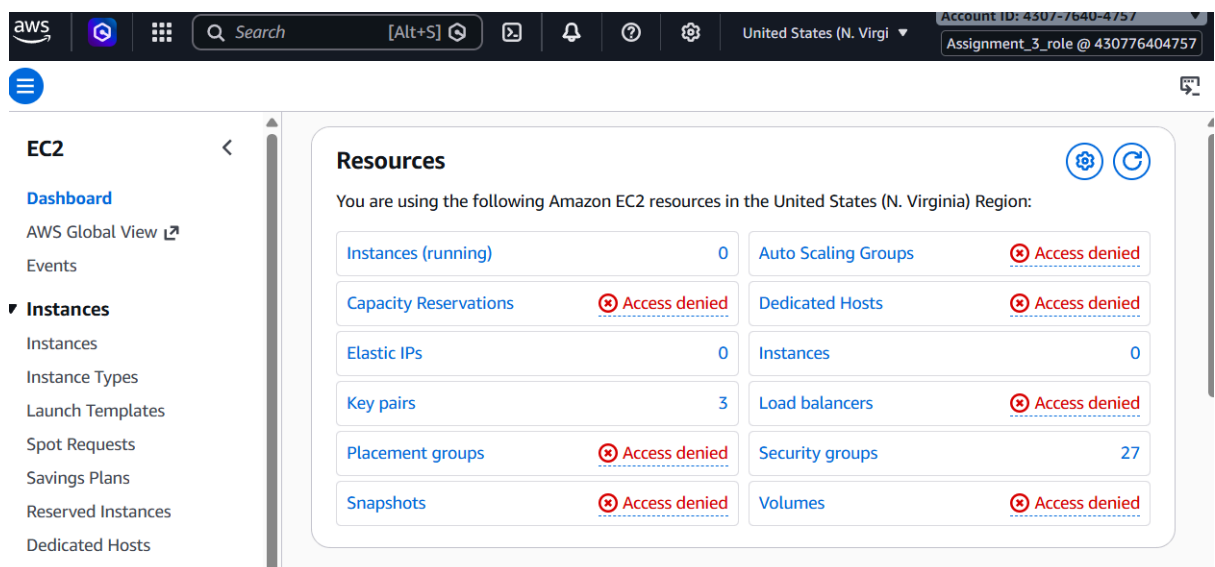


Verified Restricted Access

32. The **EC2 Dashboard** was opened

33. The available EC2 resources were reviewed.

34. Resources such as **Auto Scaling Groups, Capacity Reservations, Dedicated Hosts, Load Balancers, Placement Groups, Snapshots, and Volumes** displayed **Access denied**.



Result

The role successfully restricted access to unauthorized resources while allowing access to the services and resources permitted by its assigned policies.